



OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester 2 Examinations 2025/2026

Course Instance Codes	3BA1, 1EM1, 4BCT1, 3BMS2, 3BS9, 4BS2, 3FM2, 4FM2
Exams	3rd and 4th year (Arts and Science), Erasmus
Module Code	MA342
Module	Topology
External Examiner	Dr Michael Mackey
Internal Examiner	Dr James Cruickshank *Dr Tobias Rossmann
<u>Instructions:</u>	Answer all four questions. Each question carries 25 marks. All workings must be shown.
Duration	2 hours
No. of Pages	3 pages, including this one
Discipline	Mathematics
<u>Requirements:</u>	No special requirements
Release to Library	Yes
Release in Exam Venue	Yes
MCQ	No
Statistical / Log Tables	No
Non-programmable Calculator	Yes
Mathematical Tables	No

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- Q1.** (a) Let X be a set. Define what is meant by a *topology* on X . [4 marks]
- (b) (i) Define the *standard topology* on the set of real numbers $\mathbf{R}$.
(ii) Define the *cofinite topology* on an arbitrary set X .
(iii) Do the standard topology and the cofinite topology on $\mathbf{R}$ coincide? Justify your answer. [6 marks]
- (c) Let X be a topological space and let $A \subseteq X$ be a subset.
(i) Define the *closure* $\bar{A}$ of A .
(ii) Define the *interior* A° of A .
(iii) Define the *subspace topology* on A . [6 marks]
- (d) We endow $\mathbf{R}$ with the standard topology. Is the subspace topology on the closed interval $[0, 1]$ discrete? Justify your answer. [4 marks]
- (e) Let X be a topological space and let $A \subseteq X$. Show that if A does not have an accumulation point in X , then A is closed in X . [5 marks]
- Q2.** (a) Describe a base for the *lower limit topology* on $\mathbf{R}$. [3 marks]
- (b) (i) Let X be a topological space. Define what it means for a subset A of X to be *dense* in X .
(ii) Give an example (with brief justification) of a subset A of $\mathbf{Z}$ (the set of integers) with $A \neq \mathbf{Z}$ and such that A is dense in the cofinite topology on $\mathbf{Z}$. [6 marks]
- (c) (i) Define what it means for a topological space to be *Hausdorff*.
(ii) Let $\mathbf{R}_\ell$ denote the real line endowed with the lower limit topology. Is $\mathbf{R}_\ell$ Hausdorff? Justify your answer. [6 marks]
- (d) (i) Define what it means for a topological space to be *compact*.
(ii) Is $\mathbf{R}_\ell$ (see Q2(c)(ii)) compact? Justify your answer. [5 marks]
- (e) Let X be a Hausdorff space. Let A be a compact subspace of X . Show that A is closed in X . [5 marks]

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- Q3.** (a) Define the *final topology* on a set with respect to a family of functions. [3 marks]
- (b) Define what it means for a topological space to be *connected*. [3 marks]
- (c) For each of the following topologies $\mathcal{O}$ on the set of rational numbers $\mathbf{Q}$, decide whether the resulting topological space $(\mathbf{Q}, \mathcal{O})$ is connected or not. For disconnected spaces, exhibit an example of a separation. For connected spaces, provide a brief justification.
- (i) The discrete topology $\mathcal{O}_{\text{dis}}$.
 - (ii) The indiscrete topology $\mathcal{O}_{\text{ind}}$.
 - (iii) The subspace topology induced by the standard topology on $\mathbf{R}$.
 - (iv) The cofinite topology $\mathcal{O}_{\text{cof}}$. [10 marks]
- (d) (i) Show that $\mathbf{S}^1$ (the unit circle) and $[0, 1]$ are not homeomorphic.
- (ii) Show that $\mathbf{R}$ and $[0, 1]$ are not homeomorphic.
- (iii) Show that $\mathbf{R}$ and $(0, \infty)$ are homeomorphic. [9 marks]
- Q4.** (a) Define the *Cantor set* C . Briefly explain why C is a compact Hausdorff space. [5 marks]
- (b) (i) Let $f: X \rightarrow Y$ be a continuous surjection between topological spaces. Show that if X is connected, then so is Y .
- (ii) Briefly explain how the preceding part implies the Intermediate Value Theorem from real analysis. [6 marks]
- (c) Briefly explain why $\mathbf{S}^1$ is a quotient of $[0, 1]$. [5 marks]
- (d) Is $\mathbf{R}$ (with the standard topology) a quotient of the Cantor set C ? Justify your answer. [3 marks]
- (e) Let X and Y be topological spaces.
- (i) Let $f: X \rightarrow Y$ and $g: X \rightarrow Y$ be continuous functions. Define what it means for f and g to be *homotopic*.
 - (ii) Let $f: \mathbf{R}^n \rightarrow \mathbf{R}^n$ and $g: \mathbf{R}^n \rightarrow \mathbf{R}^n$ be continuous functions. Show that f and g are homotopic. [6 marks]